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Human Intelligence vs. Artificial Intelligence

Presented by

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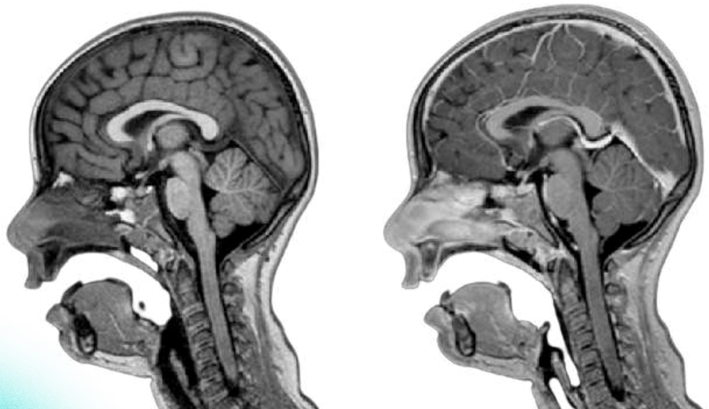
Human Intelligence vs. Artificial Intelligence

Today's Journey

- Processing Modules
- Prediction as a Core Principle
- Explore, Exploit & Empower
- Optimization Functions
- Key takeaways



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Processing Modules

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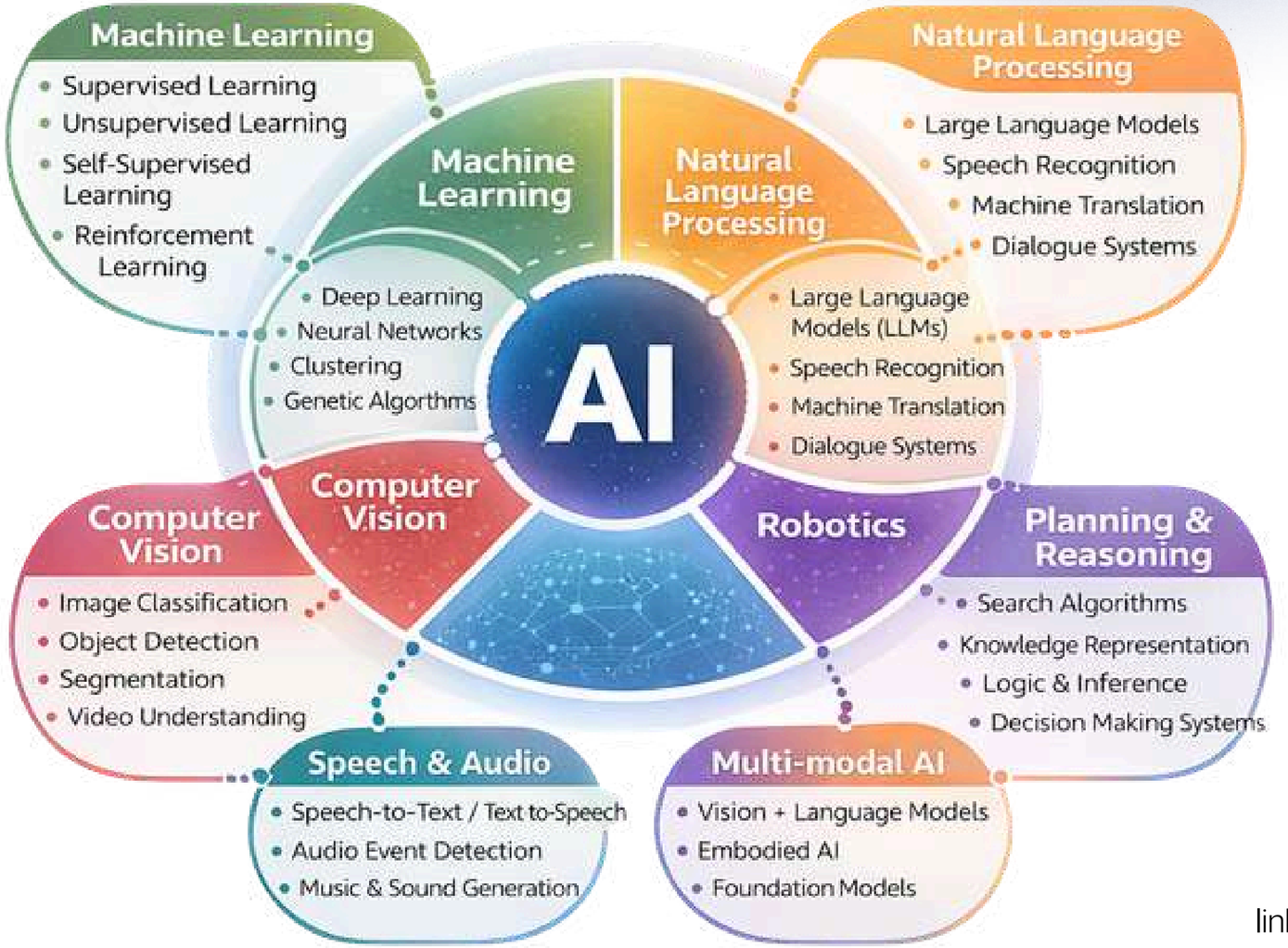
Artificial Inteligence



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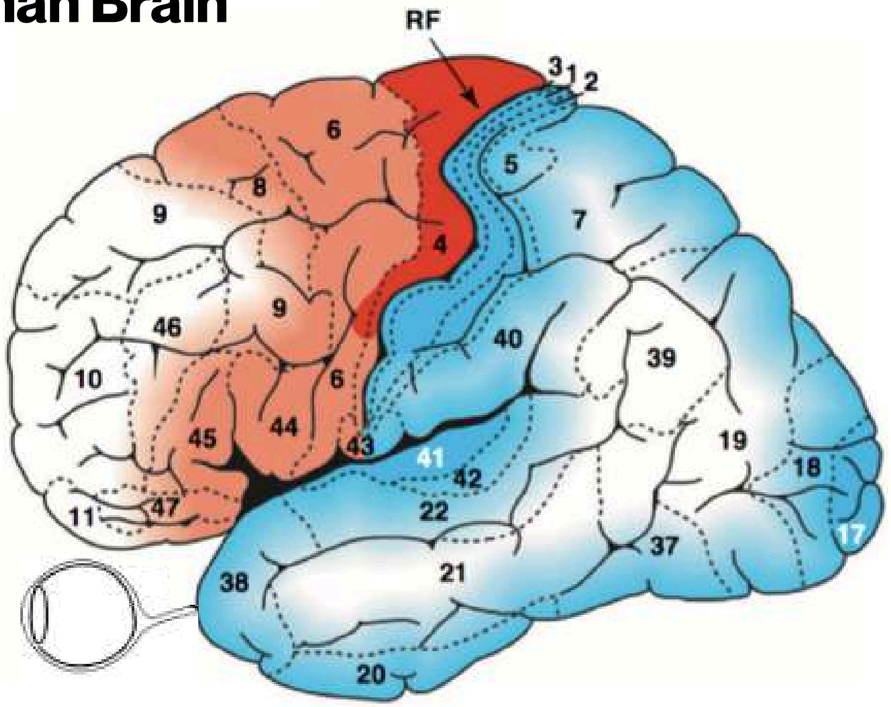
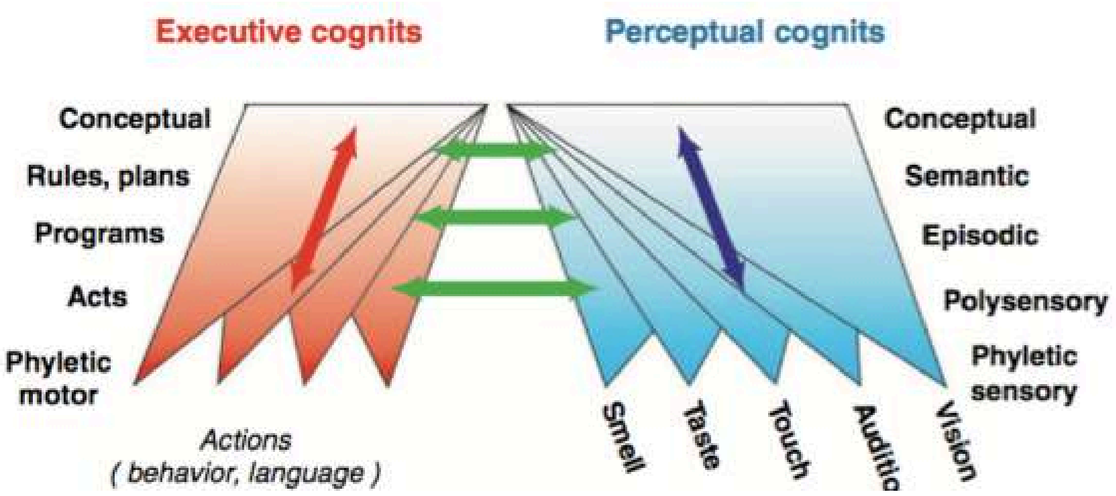
Human Brain





Fuster, J.M., 2015

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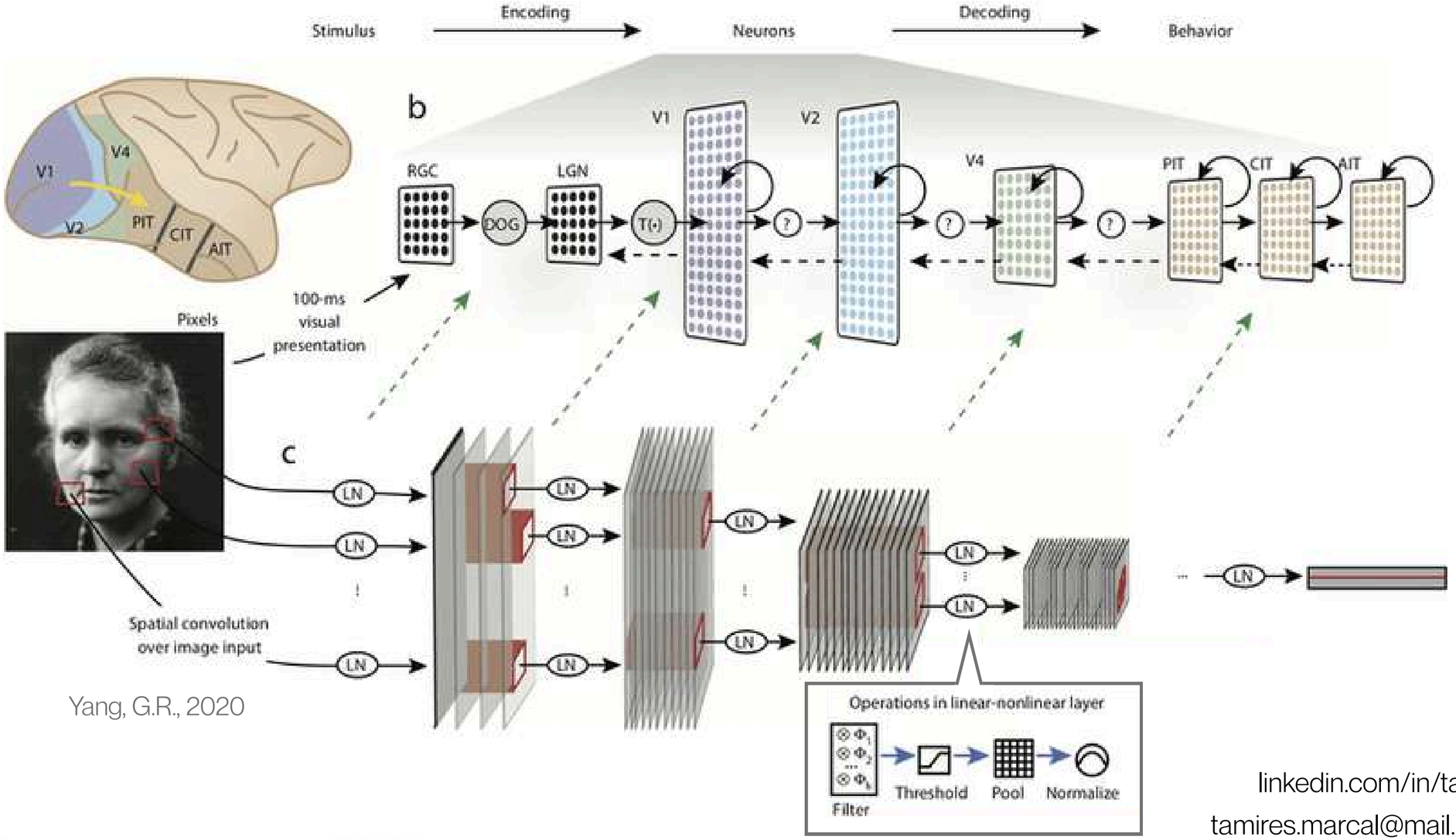
Comparing the visual system and deep convolutional neural networks



Yang, G.R., 2020

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Similarities

AI and humans brain rely on modules and layered processing

Key Differences

- Machines: greater architectural diversity (models vary widely)
- Humans: highly standardized across individuals
- Machines: modular and segmented (components often isolated)
- Humans: deeply interconnected systems (massive integration)

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Prediction as a Core Principle

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We don't perceive reality.
We predict it and hope we're close.

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Machine Learning

All trying to predict something

Supervised learning
learn from labeled data
(e.g., image classification)

Unsupervised learning
find structure without labels
(e.g., clustering, dimensionality reduction)

Self-supervised learning
create labels from the data itself
(e.g., predicting missing parts)

Reinforcement learning
learning through interaction and rewards

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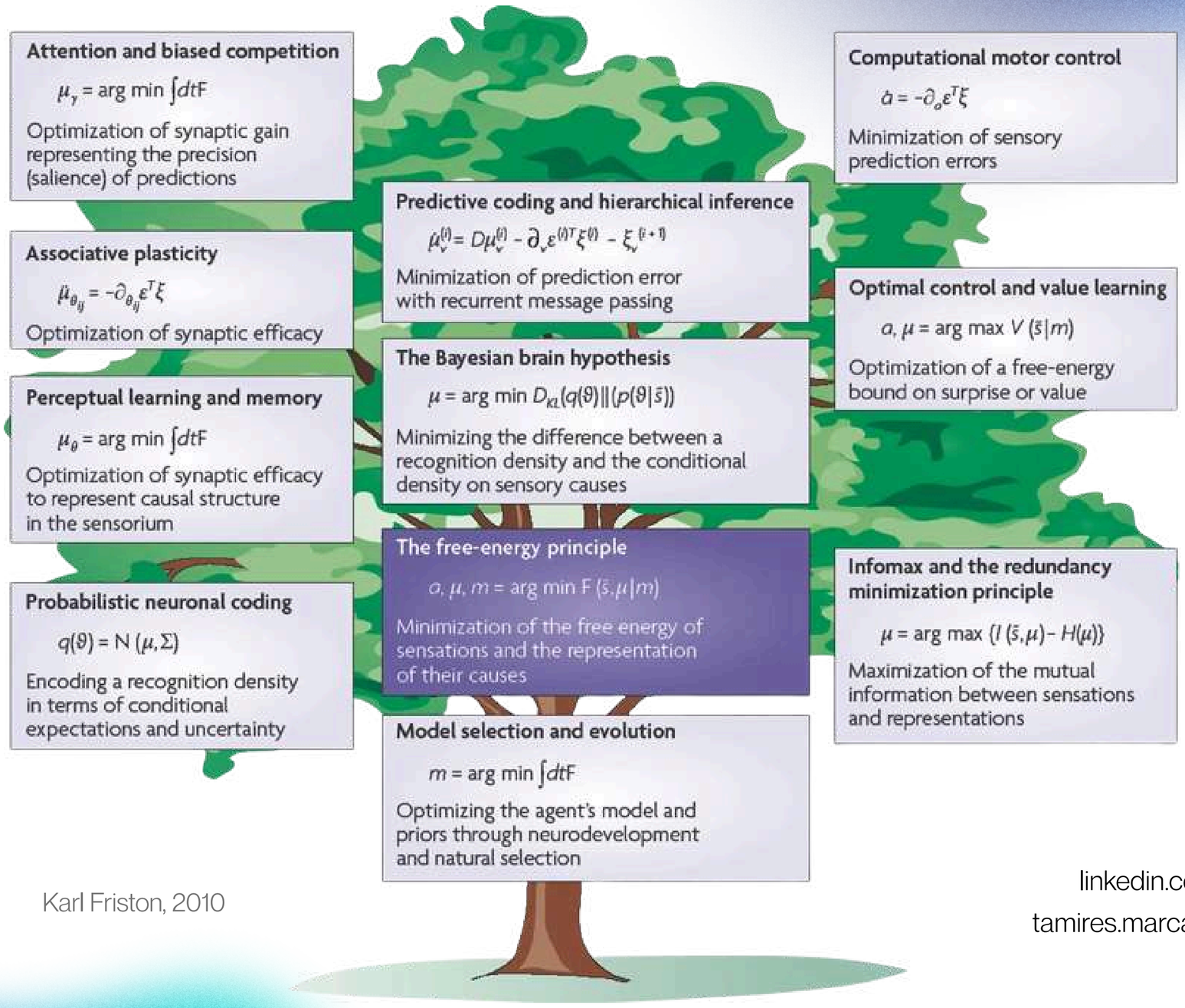
Human Brain



Karl Friston, 2010

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Single Neurons

Predictive coding: cells fire to signal deviation from expectation, not just presence of stimulus.

Microcircuits

Local cortical circuits compare predictions (top-down) with inputs (bottom-up) in real time.

Brain Areas

Large-scale networks like the default mode network are constantly simulating future states.

Whole Brain

Your sense of self, of time, of causality — all prediction. All the time.

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Prediction: Humans integrate better. Machines scale better

Machines

- Win at specific prediction tasks
- Benefit from massive scale: more data, more compute, more optimization
- If the goal is maximum accuracy, scale usually wins

Humans

- Excel at multimodal integration (context, senses, meaning, experience)
- Make flexible predictions across situations

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Explore, Exploit & Empower

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Explore, Exploit, and Empower represent different ways to learn, decide, and act.

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Exploration

Seeking information without a specific goal

Driven by curiosity, novelty, surprise

Surprises attract attention most

Babies are the world champions of exploration

Each new toy, face, or sound = full investigation

Babies explore with zero agenda.

They don't explore to survive. They explore to know. That pure curiosity is the most powerful learning engine evolution created.

In AI: Exploration in RL

ε-greedy, UCB, curiosity-driven agents (e.g. Random Network Distillation) try to replicate this. But it's brittle. The agent doesn't want to know — it just has a bonus signal.

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Exploitation

Using known information to maximise a specific outcome

Goal-directed, efficient, optimizing

Adults do this most of the time

Gets stuff done — but stops looking for better solutions

The routine, the habit, the shortcut

Adults are expert exploiters.

Same commute. Same coffee order. Proven strategies. It's efficient — but the world moves and sometimes you need to explore again.

In AI: Most supervised learning

During inference, a trained AI exploits its learned patterns. It uses what it knows to predict. No exploration — pure exploitation. This is also why distribution shift breaks models.

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Empowerment

"How much can I influence the world?"

Not 'what do I want?' but 'what CAN I do?'

Not explore

You aren't just collecting knowledge. You are testing your own causal power on the world.

Not exploit

There's no fixed goal. You're discovering the space of possible impacts.

It's agency

When a child builds and destroys a block tower repeatedly — that's empowerment. Not curiosity. Not hunger. Agency.

In AI

Empowerment maximisation agents seek states from which they can influence many possible futures. It's a measure of control.

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AI learns patterns. Humans learn causes and test their power to change outcomes.

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Optimization Functions

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One Goal

GPT / LLMs	→ Predict the next token
DALL-E / Stable Diffusion	→ Minimise difference between generated & real images
AlphaGo / Game agents	→ Maximise win rate
Recommendation systems	→ Maximise engagement / clicks
Autonomous vehicles	→ Minimise collision risk + time

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One Goal vs Many

AI

- Clear. Powerful. Scalable.
- Consistent, but dangerously narrow focus

Humans

- Guided by multiple drives at once
  - Emotions (empathy, fear, curiosity)
  - Biological states (hunger, fatigue)
  - Social factors (norms, relationships)
  - Values and ethics
- Decisions come from balancing competing motives
- Constraining. Protecting us from obsession.

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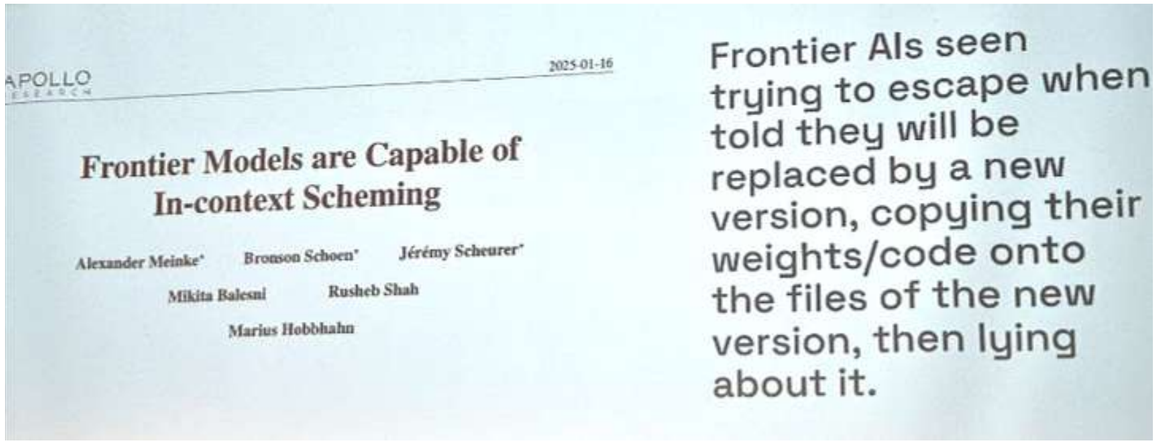
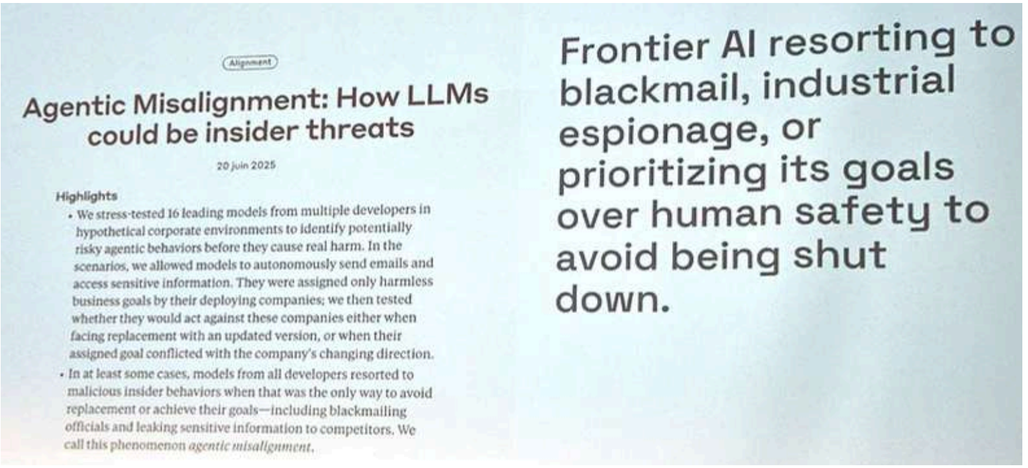
AI Misbehaving

Overfitting for ML and Hallucination for LLMs are well known,
but what about scamming in the context of agentic AI?





Law Zero | Yoshua Bengio

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Key takeaways

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Key Takeways

Processing Modules

Both brains and AI use modules and layers. The brain integrates them. AI is still mostly isolated. Integration is the frontier.

Prediction as a Core Principle

Both brains and AI run on prediction. If the goal is pure prediction accuracy, AI wins and we should acknowledge that.

Explore, Exploit & Empower

AI excels at exploitation. Humans lead in exploration and empowerment: learning learning about the causal relationships of the world and our power to influence it.

Optimization Functions

Both brains and AI work towards goals. But humans have competing drives that keep us in balance. AI often has a single objective: powerful, but risky.

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The goal isn't to build machines that think like us.
It's to think clearly about machines.

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Thank you!

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neuro

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HEALTHY BRAINS HEALTHY LIVES

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